■ Maths: Statistics

Box Plots & Interquartile Range (IQR)

Statistics helps us **collect, organise, and understand data**. In this lesson, we'll learn how to summarise data using box plots and measure spread using the IQR.

■ Key Concepts

Data Set

A collection of numbers or information gathered for analysis.

Distribution

Shows how data values spread out or cluster together.

Statistics

The study of collecting, organising, and interpreting data.

■ The Five-Number Summary

Term	Symbol	Definition
Minimum	Min	The **smallest** value in your data set
Lower Quartile	Q1	The **median of the lower half** of the data
Median	Q2	The **middle value** when data is ordered
Upper Quartile	Q3	The **median of the upper half** of the data
Maximum	Max	The **largest** value in your data set

■ Interquartile Range (IQR)

$$IQR = Q3 - Q1$$

The IQR shows the **middle 50% of your data**. A **large IQR** means data is spread out widely. A **small IQR** means data is clustered together. The IQR also helps us **spot outliers** — values unusually far from the rest.

■ Steps to Find the Five-Number Summary

1

Order the Data

Arrange all values from smallest to largest.

2

Find the Median

Identify the middle value (Q2). This splits the data into two halves.

3

Find Q1 & Q3

Q1 = median of the lower half Q3 = median of the upper half

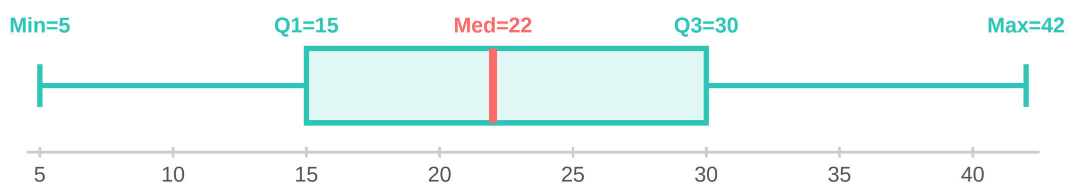
✏■ Worked Example

Find the five-number summary and IQR for: **3, 7, 8, 12, 14, 17, 21, 24, 28**

Step	Working	Result
1. Order	3, 7, 8, 12, **14**, 17, 21, 24, 28	Already ordered ✓
2. Median (Q2)	9 values → middle is the 5th value	**Q2 = 14**
3. Lower half	3, 7, **8**, 12 → median = (7+8)/2... wait, 4 values → median = (7+8)/2	**Q1 = 7.5**
4. Upper half	17, **21**, 24, 28 → median = (21+24)/2	**Q3 = 22.5**
5. IQR	IQR = Q3 – Q1 = 22.5 – 7.5	**IQR = 15**

■ Five-Number Summary: Min=3, Q1=7.5, Median=14, Q3=22.5, Max=28 | IQR = 15

■ Reading a Box Plot



■ **The Box:** Spans from Q1 to Q3, showing the middle 50% of the data (the IQR).

■ **The Median Line:** A vertical line inside the box marks the median (Q2).

■ **The Whiskers:** Lines extending left and right from the box to the Min and Max values.

■■ **Symmetrical:** Equal whiskers and a centred median line indicate symmetric data.

➡■ **Skewed:** A longer whisker on one side shows the data is skewed in that direction.

■ Practice Problems

Problem 1: Find the IQR

Data set: **1, 15, 20, 20, 21, 23, 28, 28**

Your Working	Answer
Step 1 – Order the data: ___________________________	
Step 2 – Find the Median (Q2): ___________________________	Q2 = ______
Step 3 – Find Q1 (lower half median): ___________________________	Q1 = ______
Step 4 – Find Q3 (upper half median): ___________________________	Q3 = ______
Step 5 – Calculate IQR = Q3 – Q1:	IQR = ______

Problem 2: Interpreting a Box Plot

A box plot for test scores shows: Min=50, Q1=62, Median=74, Q3=85, Max=100

a) What is the IQR?

IQR = Q3 – Q1 = _____ – _____ = _____

b) What is the range?

Range = Max – Min = _____ – _____ = _____

c) What percentage of scores are between Q1 and Q3?

____

d) Is the data skewed? How can you tell?

____

■ Comparing Two Box Plots

When you compare two box plots placed on the same **same number line**, look at:

- **Medians** — which group has the higher typical value?
- **IQR (box width)** — which group's middle 50% is more spread out?

• **Whisker length** — which group has more extreme values?
• **Overlap** — do the boxes share the same range? This shows similarity.

■ Answer Key

Problem 1 Answers	**Problem 2 Answers**
Ordered: 1, 15, 20, 20, 21, 23, 28, 28 Median (Q2) = (20+21)/2 = 20.5 Q1 = median of {1,15,20,20} = (15+20)/2 = 17.5 Q3 = median of {21,23,28,28} = (23+28)/2 = 25.5 IQR = 25.5 – 17.5 = **8**	a) IQR = 85 – 62 = **23** b) Range = 100 – 50 = **50** c) **50%** of the scores d) Slightly left-skewed (longer lower whisker)